

- Metabolized primarily by CYP450 3A4 and CYP450 3A5

Drug Interactions

- In the presence of strong/moderate CYP450 3A4 inhibitors (e.g., ketoconazole), recommended dose of pimavanserin is 10 mg once daily
- Avoid concomitant use with moderate or strong CYP450 3A4 inducers (e.g., carbamazepine)
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval

Other Warnings/Precautions

- Pimavanserin can cause prolongation of the QTc interval

Do Not Use

- If there is a proven allergy to pimavanserin

Special Populations

Renal Impairment

- Dose adjustment not necessary
- Use with caution in patients with severe impairment or end-stage renal disease

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Pimavanserin can cause QTc prolongation and should be avoided in patients with known QT prolongation or in combination with drugs that are known to prolong QT interval
- Pimavanserin should be avoided in patients with a history of cardiac arrhythmias, symptomatic bradycardia, hypokalemia, or hypomagnesemia, or presence of congenital prolongation of the QT interval

Elderly

- Dose adjustment not necessary
- Pimavanserin is not approved for the treatment of dementia-related psychosis UNRELATED to the hallucinations and delusions associated with Parkinson's disease psychosis, such as the behavioral symptoms of comorbid Alzheimer dementia

However, pimavanserin is not contraindicated for patients with dementia RELATED to Parkinson's disease who have hallucinations and delusions of Parkinson's disease psychosis



Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In rat and rabbit studies, pimavanserin did not demonstrate teratogenicity at doses up to 10 or 12 times the maximum recommended human dose
- In rat studies, doses 2 times the maximum recommended daily dose in humans based on area under the curve (AUC) resulted in maternal toxicity, including mortality and reduced body weight and food consumption, with corresponding decreases in pup survival, reduced litter size, and reduced pup body weight

Breast Feeding

- Unknown if pimavanserin is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended to either discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on pimavanserin should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Does not worsen motor symptoms of Parkinson's disease
- Not associated with the metabolic side effects of quetiapine and clozapine, including weight gain, dyslipidemia, and diabetes mellitus
- Not associated with the sedation sometimes caused by quetiapine and clozapine
- Does not require dose reduction of antiparkinsonian dopaminergic therapy given concomitantly

Potential Disadvantages

- Expensive

Primary Target Symptoms

- Hallucinations and delusions associated with Parkinson's disease
- Hallucinations and delusions associated with all causes of dementia



Pearls

- Improvement in negative symptoms of schizophrenia when augmenting atypical antipsychotics has been reported; now in phase 3 trials for negative symptoms of schizophrenia

- Some indications of improvement in depression when augmenting antidepressants, with further trials in progress
- Theoretically should be effective for the hallucinations associated with Lewy body psychosis
- Enhances slow-wave sleep and may have hypnotic properties

Suggested Reading

Cummings J , Isaacson S , Mills R , et al. Pimavanserin for patients with Parkinson's disease psychosis: a randomized, placebo-controlled phase 3 trial . *Lancet* 2014 ;**383** (9916):533–40 .

Hacksell U , Burstein ES , McFarland K , Mills RG , Williams H. On the discovery and development of pimavanserin: a novel drug candidate for Parkinson's psychosis . *Neurochem Res* 2014 Oct;**39** (10):2008–17 .

Hermanowicz S , Hermanowicz N. The safety, tolerability and efficacy of pimavanserin tartrate in the treatment of psychosis in Parkinson's disease . *Expert Rev Neurother* 2016 ;**16** (6):625–33 .

Howland RJ . Pimavanserin: an inverse agonist antipsychotic drug . *J Psychosoc Nurs Ment Health Serv* 2016 ;**54** (6):21–4 .

Stahl SM . Mechanism of action of pimavanserin in Parkinson's disease psychosis: targeting serotonin 5HT_{2A} and 5HT_{2C} receptors . *CNS Spectr* 2016 ;**21** (4):271–5 .

Pimozide

Therapeutics

Brands

- Orap
- *see index for additional brand names*

Generic?

Not in USA

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Tourette's syndrome/tic suppressant; conventional antipsychotic (neuroleptic, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Suppression of motor and phonic tics in patients with Tourette's syndrome who have failed to respond satisfactorily to standard treatment**
- Psychotic disorders in patients who have failed to respond satisfactorily to standard treatment

How the Drug Works

- Blocks dopamine 2 receptors in the nigrostriatal dopamine pathway, reducing tics in Tourette's syndrome
- When used for psychosis, can block dopamine 2 receptors in the mesolimbic dopamine pathway, reducing positive symptoms of psychosis

How Long Until It Works

- Relief from tics may occur more rapidly than antipsychotic actions
- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- * Is a second-line treatment option for Tourette's syndrome
- * Is a secondary or tertiary treatment option for psychosis or other behavioral disorders
- Should evaluate for switching to an antipsychotic with a better/risk benefit ratio

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic

- If 2 or more antipsychotic monotherapies do not work, consider clozapine

Best Augmenting Combos for Partial Response or Treatment Resistance

* Augmentation of pimozide has not been systematically studied and can be dangerous, especially with drugs that can either prolong QTc interval or raise pimozide plasma levels

Tests

- * Baseline ECG and serum potassium levels should be determined
- * Periodic evaluation of ECG and serum potassium levels, especially during dose titration
- Serum magnesium levels may also need to be monitored
- * Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including

nutrition and weight management, physical activity counseling, smoking cessation, and medical management

- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic
- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and pimozide should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia
- * Drug-induced parkinsonism

- * Tardive dyskinesia
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Hypotension
- Sedation, akinesia
- Galactorrhea, amenorrhea
- Dry mouth, constipation, blurred vision
- Sexual dysfunction
- * Weight gain



Life-Threatening or Dangerous Side Effects

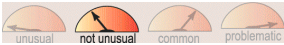
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- * Dose-dependent QTc prolongation
- Ventricular arrhythmias and sudden death
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Occurs in significant minority

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia

Best Augmenting Agents for Side Effects

* Augmentation of pimozide has not been systematically studied and can be dangerous, especially with drugs that can either prolong QTc interval or raise pimozide plasma levels

Dosing and Use

Usual Dosage Range

- Less than 10 mg/day

Dosage Forms

- Tablet 1 mg scored, 2 mg scored

How to Dose

- Initial 1–2 mg/day in divided doses; can increase dose every other day; maximum 10 mg/day or 0.2 mg/kg per day
- Children: initial 0.05 mg/kg per day at night; can increase every 3 days; maximum 10 mg/day or 0.2 mg/kg per day



Dosing Tips

* The effects of pimozide on the QTc interval are dose-dependent, so start low and go slow while carefully monitoring QTc interval

* Sudden, unexpected deaths have occurred in patients taking high doses of pimozide for conditions other than Tourette's disorder; therefore, patients should be instructed not to exceed the prescribed dose of pimozide

- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Deaths have occurred; drug-induced parkinsonism, ECG changes, hypotension, respiratory depression, coma

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration (over 6 to 8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)

- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after pimozide is discontinued

Pharmacokinetics

- Metabolized by CYP450 3A and to a lesser extent by CYP450 1A2
- Mean elimination half-life approximately 55 hours

Drug Interactions

- May decrease the effects of levodopa, dopamine agonists
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- May increase the effects of antihypertensive drugs
- * Use with CYP450 3A4 inhibitors (e.g., drugs such as fluoxetine, sertraline, fluvoxamine, and nefazodone; foods such as grapefruit juice) can raise pimozide levels and increase the risks of dangerous arrhythmias
- Use of pimozide and fluoxetine may lead to bradycardia
- Additive effects may occur if used with CNS depressants
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome

- Combined use with epinephrine may lower blood pressure

Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
 - Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
 - Antiemetic effect can mask signs of other disorders or overdose
 - Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
 - Use only with caution if at all in Parkinson's disease or Lewy body dementia
 - Because pimozide may dose-dependently prolong QTc interval, use with caution in patients who have bradycardia or who are taking drugs that can induce bradycardia (e.g., beta blockers, calcium channel blockers, clonidine, digitalis)
 - Because pimozide may dose-dependently prolong QTc interval, use with caution in patients who have hypokalemia and/or hypomagnesemia or who are taking drugs that can induce hypokalemia and/or magnesemia (e.g., diuretics, stimulant laxatives, intravenous amphotericin B, glucocorticoids, tetracosactide)
 - Pimozide can increase tumors in mice (dose-related effect)
- * Pimozide can increase the QTc interval and potentially cause arrhythmia or sudden death, especially in combination with drugs that

raise its levels

- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression
 - * If patient is taking an agent capable of significantly prolonging QTc interval (e.g., thioridazine, selected antiarrhythmics, moxifloxacin, and sparfloxacin)
 - * If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If patient is taking drugs that can cause tics
 - * If patient is taking drugs that inhibit pimozone metabolism, such as macrolide antibiotics, azole antifungal agents (ketoconazole, itraconazole), protease inhibitors, nefazodone, fluvoxamine, fluoxetine, sertraline, etc.
- If there is a proven allergy to pimozone
- If there is a known sensitivity to other antipsychotics

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Pimozide produces a dose-dependent prolongation of QTc interval, which may be enhanced by the existence of bradycardia, hypokalemia, congenital or acquired long QTc interval, which should be evaluated prior to administering pimozide
- Use with caution if treating concomitantly with a medication likely to produce prolonged bradycardia, hypokalemia, slowing of intracardiac conduction, or prolongation of the QTc interval
- Avoid pimozide in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure

Elderly

- Some patients may tolerate lower doses better
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for

treatment of elderly patients with behavioral symptoms of dementia such as agitation

- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Safety and efficacy established for patients over age 12
- Preliminary data show similar safety for patients ages 2–12 as for patients over 12
- Generally use second-line after atypical antipsychotics and other conventional antipsychotics

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Renal papillary abnormalities have been seen in rats during pregnancy

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy
- Should evaluate for an antipsychotic with a better risk/benefit ratio if treatment required during pregnancy

Breast Feeding

- Unknown if pimozide is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
 - Not recommended for use because of potential for tumorigenicity or cardiovascular effects on infant
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Only for patients who respond to this agent and not to other antipsychotics

Potential Disadvantages

- Vulnerable populations such as children and elderly
- Patients on other drugs

Primary Target Symptoms

- Vocal and motor tics in patients who fail to respond to treatment with other antipsychotics
- Psychotic symptoms in patients who fail to respond to treatment with other antipsychotics

Pearls

* In the past, pimozide was a first-line choice for Tourette's syndrome and for certain behavioral disorders, including monosymptomatic hypochondriasis; however, it is now recognized that the benefits of pimozide generally do not outweigh its risks in most patients

* Because of its effects on the QTc interval, pimozide is not intended for use unless other options for tic disorders (or psychotic disorders) have failed

Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

.....

Pringsheim T , Marras C . Pimozide for tics in Tourette’s syndrome . *Cochrane Database Syst Rev* 2009 ;**15** (2):CD006996 .

.....

Rathbone J , McMonagle T . Pimozide for schizophrenia or related psychoses . *Cochrane Database Syst Rev* 2007 ;**18** (3):CD001949 .

Pipothiazine

Therapeutics

Brands

- Piportil
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- Maintenance treatment of schizophrenia
- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood

stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics (risperidone, olanzapine, quetiapine, ziprasidone, aripiprazole, paliperidone, amisulpride, asenapine, iloperidone, lurasidone)
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

* Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine